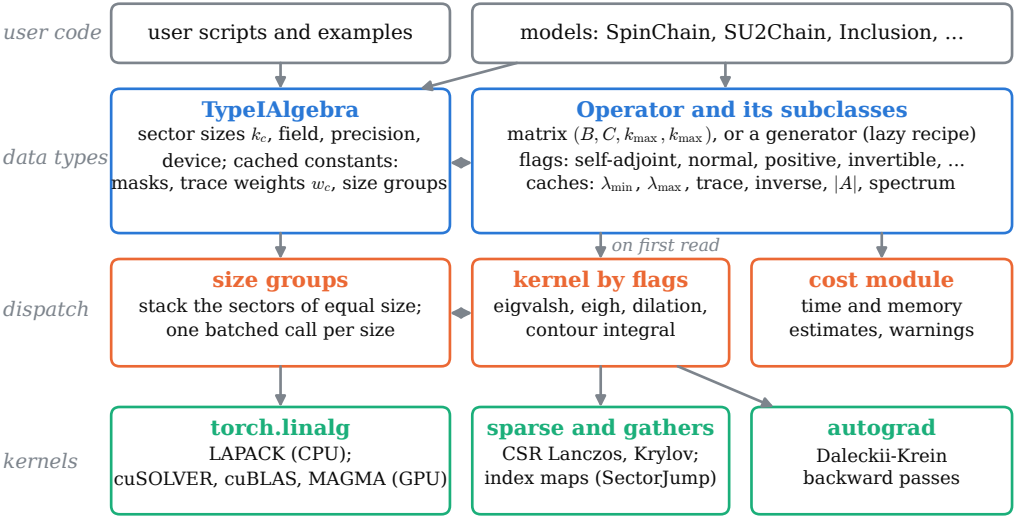
user code
user scripts and examples
models: SpinChain, SU2Chain, Inclusion, ...
data types
TypeIAlgebra
sector sizes k_c, field, precision,
device; cached constants:
masks, trace weights w_c, size groups
Operator and its subclasses
matrix $(B, C, k_{\max}, k_{\max})$, or a generator (lazy recipe)
flags: self-adjoint, normal, positive, invertible, ...
caches: $\lambda_{\min}$, $\lambda_{\max}$, trace, inverse, $|A|$, spectrum
on first read
dispatch
size groups
stack the sectors of equal size;
one batched call per size
kernel by flags
eigvalsh, eigh, dilation,
contour integral
cost module
time and memory
estimates, warnings
kernels
torch.linalg
LAPACK (CPU);
cuSOLVER, cuBLAS, MAGMA (GPU)
sparse and gathers
CSR Lanczos, Krylov;
index maps (SectorJump)
autograd
Daleckii-Krein
backward passes